

Q.1 what are the different protocols you observe at the following layers of the protocol stack?

Application Layer	Transport Layer	Network Layer
MDNS	UDP	ARP
NBNS	Tcp	ICMPV6
DHCP		IGMPV3
DNS		
HTTP		
SSDP		
DHCPV6		

Q.2 what is the total amount of data being received for the following two cases?

A. when you access <http://neverssl.com/>

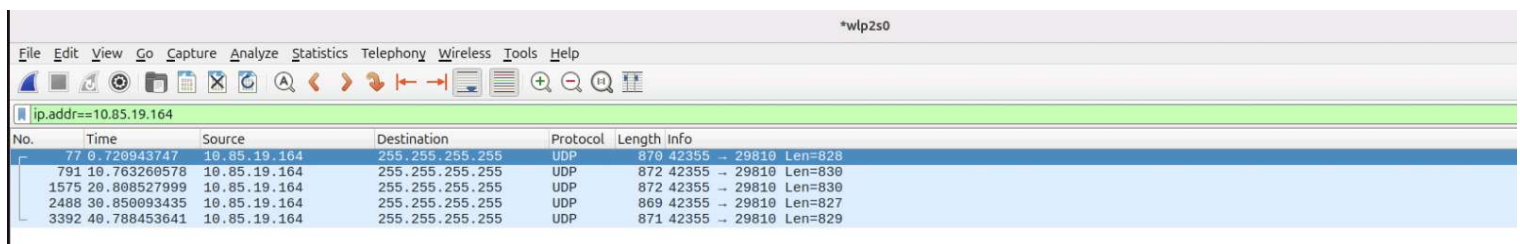
B. when you access <http://www.cornell.edu>

*wlp2s0

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ip.addr==10.85.22.209

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	10.85.22.209	255.255.255.255	UDP	867	38855 → 29810 Len=825
923	20.071120139	10.85.22.209	255.255.255.255	UDP	867	38855 → 29810 Len=825
1465	30.106658544	10.85.22.209	255.255.255.255	UDP	867	38855 → 29810 Len=825
2204	50.177054666	10.85.22.209	255.255.255.255	UDP	867	38855 → 29810 Len=825
2712	60.212571388	10.85.22.209	255.255.255.255	UDP	867	33501 → 29810 Len=825
3169	70.248383354	10.85.22.209	255.255.255.255	UDP	867	33501 → 29810 Len=825
3867	90.216083792	10.85.22.209	255.255.255.255	UDP	867	33501 → 29810 Len=825
4140	100.251148929	10.85.22.209	255.255.255.255	UDP	867	33501 → 29810 Len=825
4758	110.287437067	10.85.22.209	255.255.255.255	UDP	867	33501 → 29810 Len=825
5299	120.319921980	10.85.22.209	255.255.255.255	UDP	867	33501 → 29810 Len=825
6015	130.357438927	10.85.22.209	255.255.255.255	UDP	867	33501 → 29810 Len=825
6880	140.390127437	10.85.22.209	255.255.255.255	UDP	867	33501 → 29810 Len=825



Q.3 Answer the following when you access the site: //neverSSL.com/.

A.) How many HTTP GET requests do you observe?

One HTTP GET request observed.

GET requests:-

Time	Source	Destination	Protocol	Length	Info
22.733866912	10.85.16.204	91.189.91.96	HTTP	154	GET/HTTP/1.1

B) What version of HTTP the server is running?

GET/HTTP/1.1

C) For each of the HTTP GET request find out ;

i) The total number of TCP segments being received.

Ans

one TCP segment received

ii) The total amount of data being received in the corresponding HTTP response message.

Ans 1365 bytes.

Transmission Control Protocol, Src Port: 80, Dst Port: 57952, Seq: 2897, Ack: 76, Len: 1365

Source Port: 80

Destination Port: 57952

[Stream index: 2]

[Conversation completeness: Complete, WITH_DATA (31)]

[TCP Segment Len: 1365]

Sequence Number: 2897 (relative sequence number)

Sequence Number (raw): 982771459

[Next Sequence Number: 4262 (relative sequence number)]

Acknowledgment Number: 76 (relative ack number)

Acknowledgment number (raw): 2748014050

1000 = Header Length: 32 bytes (8)

Flags: 0x018 (PSH, ACK)

Window: 210

[Calculated window size: 26880]

[Window size scaling factor: 128]

Checksum: 0xc5d1 [unverified]

[Checksum Status: Unverified]

Urgent Pointer: 0

Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps

[Timestamps]

[SEQ/ACK analysis]

TCP payload (1365 bytes)

Hypertext Transfer Protocol

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0020  10 cc 00 50 e2 60 3a 93 e7 03 a3 cb 5d e2 80 18 ...P...].
0030  00 d2 c5 d1 00 00 01 01 08 0a cc 63 fa 97 10 3e .....c...>
0040  07 91 77 6f 72 6b 2c 20 61 6e 64 20 6e 6f 74 68 ..work, and noth
0050  69 6e 67 20 68 61 70 70 65 6e 73 2e 20 54 79 70 ing happ ens. Typ
0060  65 20 22 68 74 74 70 3a 2f 2f 6e 65 76 65 72 73 e "http: //nevers
0070  73 6c 2e 63 6f 6d 22 0a 09 09 69 6e 74 6f 20 79 sl.com". ..into y
0080  6f 75 72 20 62 72 6f 77 73 65 72 27 73 20 75 72 our brow ser's ur
0090  6c 20 62 61 72 2c 20 61 6e 64 20 79 6f 75 27 6c l bar, a nd you'l
00a0  6c 20 62 65 20 61 62 6c 65 20 74 6f 20 6c 6f 67 l be abl e to log
00b0  20 6f 6e 2e 3c 2f 70 3e 0a 0a 09 09 3c 68 32 3e on.</p> ....<h2>
00c0  48 6f 77 3f 3c 2f 68 32 3e 0a 09 09 3c 70 3e 6e How?</h2 >...<p>n
00d0  65 76 65 72 73 73 6c 2e 63 6f 6d 20 77 69 6c 6c everssl. com will
00e0  20 6e 65 76 65 72 20 75 73 65 20 53 53 4c 20 28 never u se SSL (
00f0  61 6c 73 6f 20 6b 6e 6f 77 6e 20 61 73 20 54 4c also kno wn as TL
0100  53 29 2e 20 4e 6f 0a 09 09 65 6e 63 72 79 70 74 S). No.. -encrypt
0110  69 6f 6e 2c 20 6e 6f 20 73 74 72 6f 6e 67 20 61 ion, no strong a
0120  75 74 68 65 6e 74 69 63 61 74 69 6f 6e 2c 20 6e uthentic ation, n
0130  6f 20 3c 61 0a 09 09 68 72 65 66 3d 22 68 74 74 o <a...h ref="htt
0140  70 73 3a 2f 2f 65 6e 2e 77 69 6b 69 70 65 64 69 ps://en. wikipedi
0150  61 2e 6f 72 67 2f 77 69 6b 69 2f 48 54 54 50 5f a.org/wi ki/HTTP_
0160  53 74 72 69 63 74 5f 54 72 61 6e 73 70 6f 72 74 Strict_T ransport
0170  5f 53 65 63 75 72 69 74 79 22 3e 48 53 54 53 3c _Securit y">HSTS<
0180  2f 61 3e 2c 0a 09 09 6e 6f 20 48 54 54 50 2f 32 /a>, ...n o HTTP/2
0190  2e 30 2c 20 6a 75 73 74 20 70 6c 61 69 6e 20 6f .0, just plain o
01a0  6c 64 20 75 6e 65 6e 63 72 79 70 74 65 64 20 48 ld unenc rypted H
01b0  54 54 50 20 61 6e 64 20 66 6f 72 65 76 65 72 20 TTP and forever
01c0  73 74 75 63 6b 20 69 6e 20 74 68 65 20 64 61 72 stuck in the dar
01d0  6b 0a 09 09 61 67 65 73 20 6f 66 20 69 6e 74 65 k...ages of inte

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D) Identify TCP three way handshake used to establish the connection with <http://www.evossi.com>. Specify SYN, SYN-ACK, ACK packets and write the sequence number for each of the three accesses.

- SYN Packet:- 10.85.17.18 (Client)

Source IP:- IP: 34.223.124.45 (Server)

Destination IP:- 57022 → 80 [SYN]

Protocol info:- 0

Sequence Number:-

- SYN-ACK Packet:-

Source IP:- 34.223.124.45 (Server)

Destination IP:- 10.85.17.18 (Client)

Protocol info:- 57022 [SYN, ACK]

Sequence Number:- 0
(ACK=1)

- ACK Packet:-

Source IP:- 10.85.17.18 (Client)

Destination IP:- 34.223.124.45 (Server)

Protocol info:- 80 [ACK]

Sequence Number:- 1
(ACK=1)

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tcp

No.	Time	Source	Destination	Protocol	Length	Info
52	0.588341247	10.85.16.204	57.144.143.32	TLSv1.2	136	Application Data
78	0.725493277	57.144.143.32	10.85.16.204	TCP	66	443 → 44822 [ACK] Seq=1 Ack=71 Win=767 Len=0 TSval=3116206031 TSecr=302746751
89	0.923133150	57.144.143.32	10.85.16.204	TLSv1.2	138	Application Data
90	0.923183337	10.85.16.204	57.144.143.32	TCP	66	44822 → 443 [ACK] Seq=71 Ack=73 Win=64 Len=0 TSval=302747086 TSecr=3116206241
1584	19.970619533	10.85.16.204	57.144.143.32	TLSv1.2	136	Application Data
1592	20.090539694	57.144.143.32	10.85.16.204	TCP	66	443 → 44822 [ACK] Seq=73 Ack=141 Win=767 Len=0 TSval=3116225412 TSecr=302766133
1603	20.199042209	57.144.143.32	10.85.16.204	TLSv1.2	138	Application Data
1604	20.199087138	10.85.16.204	57.144.143.32	TCP	66	44822 → 443 [ACK] Seq=141 Ack=145 Win=64 Len=0 TSval=302766362 TSecr=3116225622
2216	33.397327759	10.85.16.204	34.223.124.45	TCP	74	46586 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=2752649712 TSecr=0 WS=1024
2265	34.121718890	10.85.16.204	57.144.143.32	TLSv1.2	185	Application Data
2273	34.367493573	57.144.143.32	10.85.16.204	TCP	66	443 → 52272 [ACK] Seq=1 Ack=40 Win=364 Len=0 TSval=3267011579 TSecr=302780284
2274	34.368704359	57.144.143.32	10.85.16.204	TLSv1.2	185	Application Data
2275	34.368763551	10.85.16.204	57.144.143.32	TCP	66	52272 → 443 [ACK] Seq=40 Ack=40 Win=63 Len=0 TSval=302780471 TSecr=3267011579
2287	34.412040522	10.85.16.204	34.223.124.45	TCP	74	[TCP Retransmission] 46586 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=2752650727 TSecr=0 WS=1024
2305	34.717187114	34.223.124.45	10.85.16.204	TCP	74	80 → 46586 [SYN, ACK] Seq=0 Ack=1 Win=26847 Len=0 MSS=1460 SACK_PERM TSval=3535133626 TSecr=2752650727
2306	34.717252472	10.85.16.204	34.223.124.45	TCP	66	46586 → 80 [ACK] Seq=1 Ack=1 Win=64512 Len=0 TSval=2752651032 TSecr=3535133626
2307	34.717441755	10.85.16.204	34.223.124.45	HTTP	141	GET / HTTP/1.1
2340	35.126670305	34.223.124.45	10.85.16.204	TCP	66	80 → 46586 [ACK] Seq=1 Ack=76 Win=29312 Len=0 TSval=3535133993 TSecr=2752651032
2341	35.12695108	34.223.124.45	10.85.16.204	TCP	66	[TCP Dup ACK 2340#1] 80 → 46586 [ACK] Seq=1 Ack=76 Win=29312 Len=0 TSval=3535133993 TSecr=2752651032
2342	35.129326800	34.223.124.45	10.85.16.204	TCP	1514	80 → 46586 [ACK] Seq=1 Ack=76 Win=29312 Len=1448 TSval=3535133994 TSecr=2752651032 [TCP segment of a r
2343	35.129372296	10.85.16.204	34.223.124.45	TCP	66	46586 → 80 [ACK] Seq=76 Ack=1449 Win=67584 Len=0 TSval=2752651444 TSecr=3535133994
2344	35.129711565	34.223.124.45	10.85.16.204	TCP	1514	80 → 46586 [ACK] Seq=1449 Ack=76 Win=29312 Len=1448 TSval=3535133994 TSecr=2752651032 [TCP segment of
2345	35.129746410	10.85.16.204	34.223.124.45	TCP	66	46586 → 80 [ACK] Seq=76 Ack=2897 Win=67584 Len=0 TSval=2752651444 TSecr=3535133994
2346	35.130062317	34.223.124.45	10.85.16.204	HTTP	1431	HTTP/1.1 200 OK (text/html)
2347	35.130096710	10.85.16.204	34.223.124.45	TCP	66	46586 → 80 [ACK] Seq=76 Ack=4262 Win=67584 Len=0 TSval=2752651445 TSecr=3535133994
2348	35.130607288	10.85.16.204	34.223.124.45	TCP	66	46586 → 80 [FIN, ACK] Seq=76 Ack=4262 Win=67584 Len=0 TSval=2752651445 TSecr=3535133994
2360	35.432134602	34.223.124.45	10.85.16.204	TCP	66	80 → 46586 [FIN, ACK] Seq=4262 Ack=77 Win=29312 Len=0 TSval=3535134410 TSecr=2752651445
2361	35.432181288	10.85.16.204	34.223.124.45	TCP	66	46586 → 80 [ACK] Seq=77 Ack=4263 Win=67584 Len=0 TSval=2752651747 TSecr=3535134410